

Math 2551 Worksheet: Scalar Line Integrals

1. For each curve, find a parameterization of the curve with the specified orientation.
 - (a) The line segment in $\mathbb{R}^3$ from $(0, 1, -2)$ to $(3, -1, 2)$.
 - (b) The line segment in $\mathbb{R}^3$ from $(3, -1, 2)$ to $(0, 1, -2)$.
 - (c) The circle of radius 3 in $\mathbb{R}^2$ centered at the origin, beginning at the point $(0, -3)$ and proceeding clockwise around the circle.
 - (d) In $\mathbb{R}^2$, the portion of the parabola $y^2 = x$ from the point $(4, 2)$ to the point $(1, -1)$.

For problems 2 to 4 below, do the set up of each line integral before doing any computations.

2. Find the line integral of $f(x, y, z) := \sqrt{x^2 + y^2}$ over the curve $\mathbf{r}(t) = (-4 \sin t)\mathbf{i} + (4 \cos t)\mathbf{j} + 3t\mathbf{k}$, $t \in [0, 2\pi]$.
3. Find the line integral of $f(x, y) = \sqrt{4x + 1}$ over C where C is the part of the curve $x = y^2$ from the point $(4, -2)$ to $(1, 1)$.
4. Let C be the curve with parameterization

$$\mathbf{r}(t) = (e^t \cos t)\mathbf{i} + (e^t \sin t)\mathbf{j} + e^t\mathbf{k}, \quad t \in [0, \pi].$$

Find the mass of C if the density of a wire along C is $\delta(x, y, z) = z^{-1}$.

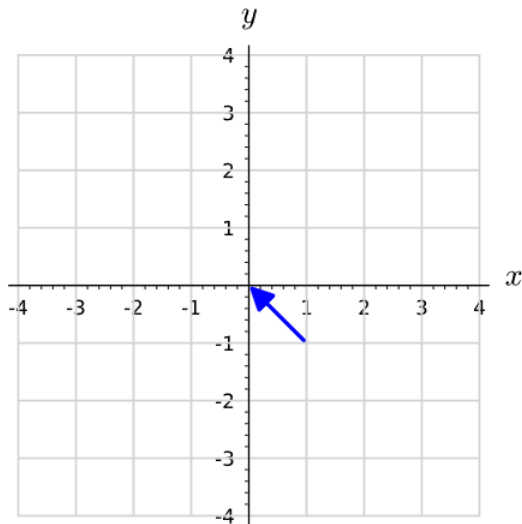
Vector Field Problems: In Chapter 13, we talked about vector-valued functions: functions whose input is a single real number and whose output is a vector in $\mathbb{R}^2$ or $\mathbb{R}^3$. A very important related type of function in this unit is a **vector field**, which is a function that takes a point (x, y) in $\mathbb{R}^2$ and outputs a vector in $\mathbb{R}^2$ or a point (x, y, z) in $\mathbb{R}^3$ and outputs a vector in $\mathbb{R}^3$.

5. If we want to make a decision based on what the wind is doing, then we need to keep track of not just its strength at any point, but also its direction! So a vector field is a good model here: at each point (x, y) , we can record the velocity vector for the wind.

Suppose that given the point (x, y) in the plane, we know that the wind velocity at that point is given by the vector $\mathbf{F}(x, y) = \langle y, x \rangle$. For example, at the point $(1, -1)$, the wind velocity is $\mathbf{F}(1, -1) = \langle -1, 1 \rangle$. Fill in the table below with the wind velocity vectors for the given points.

(x, y)	$(-2, 0)$	$(-1, 2)$	$(0, -2)$	$(1, 1)$	$(2, 3)$	$(3, 2)$	$(-1, 0)$	$(1, 3)$
$\mathbf{F}(x, y)$								

6. Another useful way of recording this information is to plot the velocity vectors! At each point (x, y) we will draw the vector $\mathbf{F}(x, y)$ starting with the tail of the vector at (x, y) . This has been done for you with the point $(1, -1)$ below. Fill in the plot with the other vectors you found in the last problem.



CalcPlot3d can graph vector fields too! Use it to check your work and see the full detail of this vector field.

7. Let $f(x, y) = \frac{1}{2}(x - y)^2$. Find the gradient vector field, ∇f , of f and sketch it.